

UNIT-5

Theoretical Continuous Distributions

Uniform Distribution/Rectangular Distribution

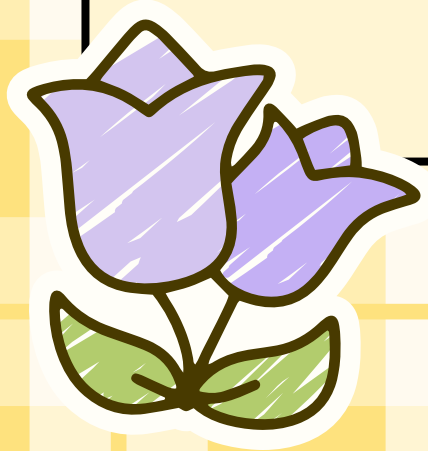
A continuous random variable assumes infinite number of values. If the values appear equally likely then the continuous random variable may be assumed to follow rectangular distribution

Notation: $X \sim U(a, b)$

For a continuous uniform distribution

$$\text{Mean} = \frac{b+a}{2}$$

$$\text{Variance} = \frac{(b-a)^2}{12}$$



Uniform Distribution/Rectangular Distribution

A random variable X is said to have a continuous uniform distribution over an interval (a, b) if its probability density function is constant k over the entire range of X ,

i.e.

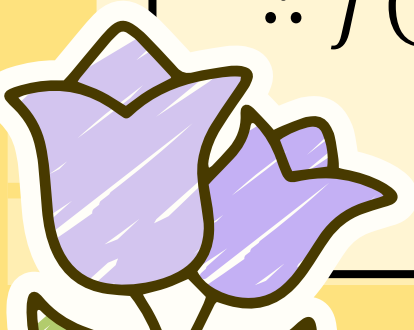
$$f(x) = k, \quad a < x < b$$
$$0, \quad \text{otherwise}$$

Since total probability is always unity

$$\int_a^b f(x) = k \int_a^b dx = 1$$

$$\Rightarrow k(b - a) = 1 \Rightarrow k = \frac{1}{b-a}$$

$$\therefore f(x) = \frac{1}{b-a}, \quad a < x < b$$
$$= 0, \text{ otherwise}$$



Examples

1. If X is uniformly distributed with mean 1 and variance $\frac{4}{3}$. Find $P(X > 0)$

$$\therefore X \sim U(a, b)$$

$$\text{Mean} = \frac{a+b}{2} = 1$$

$$\Rightarrow a+b=2 \quad \text{--- (1)}$$

$$\text{Variance} = \frac{(b-a)^2}{12} = \frac{4}{3}$$

$$(b-a)^2 = \frac{4}{3} \times 12$$

$$(b-a)^2 = 16$$

$$\text{i.e. } (b-a) = \pm 4$$

$$\text{or } b-a = +4 \quad \text{--- (2)} \quad \text{or } b-a = -4 \quad \text{--- (3)}$$

from (1) and (2)

$$a+b=2$$

$$b-a=4$$

$$2b=6$$

$$b=3$$

Put $b=3$ in (1)

$$a+b=2$$

$$a+3=2$$

$$a=-1$$

Now, from (1) and (3)

$$b+a=2$$

$$b-a=-4$$

$$2b=-2$$

$$b=-1$$

Put $b=-1$ in (1)

$$a+b=2$$

$$a=-1$$

$\therefore a=-1$ and $b=3$ OR $a=3$ and $b=-1$

$$a < b$$

$\therefore$ We consider $a=-1$ and $b=3$

SOLUTION

$$X \sim U(-1, 3)$$

$$f(x) = \frac{1}{3 - (-1)} = \frac{1}{4} \quad -1 < x < 3$$

$$\therefore f(x) = \frac{1}{4} \quad -1 < x < 3$$
$$= 0 \quad \text{o.w.}$$

$$P(X > 0) = \int_0^3 f(x) dx$$

$$= \int_0^3 \frac{1}{4} dx$$

$$= \frac{1}{4} [x]_0^3$$

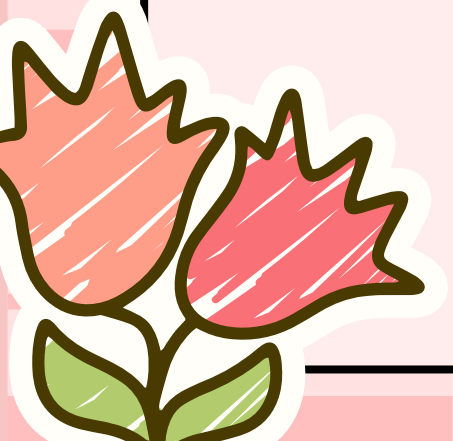
$$= \frac{1}{4} [3 - 0]$$

$$= \frac{3}{4}$$

Examples

Q.2 The daily sales of milk at a booth lies between 200 litres to 250 litres and follows uniform distribution,

- i. What is the average sale of milk on a particular day?
- ii. What is the chance that sales will exceed the average sales tomorrow?



SOLUTION

Q.2 X : Daily Sales of milk

$$X \sim U(200, 250)$$

$$f(x) = \frac{1}{250-200} = \frac{1}{50} \quad 200 < x < 250$$
$$= 0 \quad \text{o.w}$$

i) Average Sale of milk

$$X \sim U(200, 250)$$

$$f(x) = \frac{1}{50} \quad 200 < x < 250$$
$$= 0 \quad \text{o.w}$$

$$\therefore \text{Mean} = \frac{a+b}{2} = \frac{200+250}{2}$$
$$= \frac{450}{2}$$
$$= 225$$

ii) P(Sales will exceed average sales tomorrow)

$$\Rightarrow P(X > 225)$$

$$= \int_{225}^{250} f(x) dx = \int_{225}^{250} \frac{1}{50} dx$$

$$= \frac{1}{50} \left[x \right]_{225}^{250}$$

$$= \frac{1}{50} [250 - 225]$$

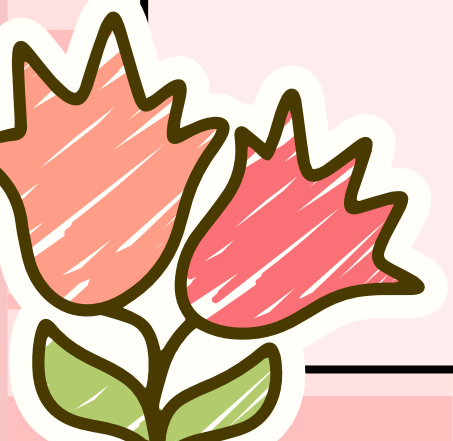
$$= \frac{25}{50}$$

$$= \frac{1}{2}$$

Examples

Q.3 The waiting time at a bustop is assumed to follow rectangular distribution over $(5,15)$. What is the chance that a person arriving at a bustop gets bus

- i. Within 10 minutes
- ii. After 7 minutes
- iii. Waiting time for the person at a bustop lies between 8 and 12 minutes



SOLUTION

3. X : Waiting time in a bustop
 $X \sim U(5, 15)$

$$f(x) = \frac{1}{10} \quad 5 < x < 15$$
$$= 0 \quad \text{o.w.}$$

i) $P(\text{Within 10 minutes})$

$$= P(X < 10)$$

$$= \int_5^{10} f(x) dx$$

$$= \int_5^{10} \frac{1}{10} dx$$

$$= \frac{1}{10} [x]_5^{10}$$

$$= \frac{1}{10} [10 - 5]$$

$$= \frac{5}{10}$$

$$= \frac{1}{2}$$

ii) After 7 mins

$$P(\text{After 7 mins}) = P(X > 7)$$

$$= \int_7^{15} f(x) dx$$

$$= \int_7^{15} \frac{1}{10} dx$$

$$= \frac{1}{10} [x]_7^{15}$$

$$= \frac{1}{10} [15 - 7]$$

$$= \frac{8}{10}$$

$$= 0.8$$

iii) Waiting time lies between 8 and 12 minutes

$$P(8 < X < 12) = \int_8^{12} f(x) dx$$

$$= \int_8^{12} \frac{1}{10} dx$$

$$= \frac{1}{10} [x]_8^{12}$$

$$= \frac{1}{10} [12 - 8]$$

$$= 0.4$$

Examples

Q.4 $X \sim U(-3,3)$. Obtain mean and variance of X . Find $P(X > 2)$ $P(X < 1)$ and K such that $P(X < K) = 0.6$

$$4. X \sim U(-3, 3)$$

$$f(x) = \frac{1}{6} \quad -3 < x < 3$$
$$= 0 \quad \text{o.w.}$$

$$\text{i) Mean} = \frac{a+b}{2}$$
$$= \frac{-3+3}{2}$$
$$= 0$$

$$\text{ii) Variance} = \frac{(b-a)^2}{12}$$
$$= \frac{[3 - (-3)]^2}{12}$$
$$= \frac{(3+3)^2}{12}$$
$$= \frac{36}{12}$$
$$= 3$$

$$\text{iii) } P(X > 2) = \int_2^3 f(x) dx$$
$$= \int_2^3 \frac{1}{6} dx$$
$$= \frac{1}{6} [x]_2^3$$
$$= \frac{1}{6} [3 - 2]$$

$$P(X > 2) = \frac{1}{6}$$

$$\text{iv) } P(X < 1) = \int_{-3}^1 f(x) dx$$
$$= \int_{-3}^1 \frac{1}{6} dx$$
$$= \frac{1}{6} [x]_{-3}^1$$
$$= \frac{1}{6} [1 - (-3)]$$
$$P(X < 1) = \frac{4}{6}$$

SOLUTION

$$P(X < k) = 0.6$$

$$\int_{-3}^k f(x) dx = 0.6$$

$$\int_{-3}^k \frac{1}{6} dx = 0.6$$

$$\frac{1}{6} [x]_{-3}^k = 0.6$$

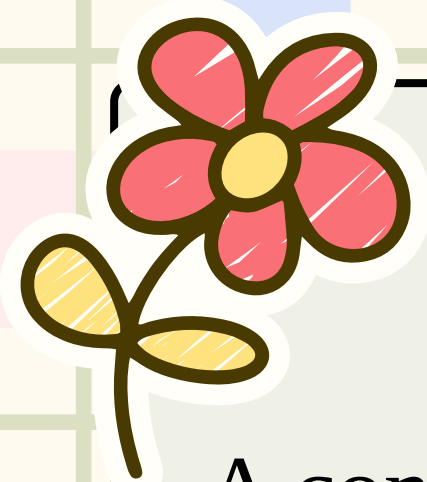
$$\frac{1}{6} [k - (-3)] = 0.6$$

$$\frac{1}{6} (k + 3) = 0.6$$

$$k + 3 = 3.6$$

$$k = 3.6 - 3$$

$$k = 0.6$$



Exponential Distribution



A continuous random variable X is said to follow exponential distribution if its pdf is given by

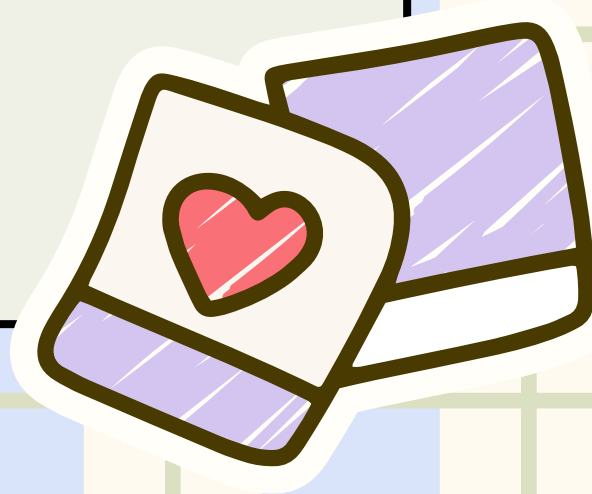
$$f(x) = \lambda \cdot e^{-\lambda x}, x > 0 \\ = 0, \text{otherwise}$$

Where $\lambda > 0$ is called the rate of the distribution.

Notation: $X \sim \exp(\lambda)$

Mean $E(X) = \frac{1}{\lambda}$

Variance $V(X) = \frac{1}{\lambda^2}$



QUESTIONS

A random variable X follows exponential distribution with mean 2.

Find $P(X < 1)$

$P(1 < X < 2)$

$P(X > 2)$

SOLUTION

$$1. X \sim \exp(1)$$

$$f(x) = \begin{cases} 1e^{-1x} & x > 0 \\ 0 & \text{o.w} \end{cases}$$

$$\therefore \text{Mean} = 2 = \frac{1}{\lambda}$$

$$\therefore \lambda = \frac{1}{2}$$

$$\therefore X \sim \exp(\lambda = \frac{1}{2})$$

$$f(x) = \begin{cases} \frac{1}{2} e^{-\frac{1}{2}x} & x > 0 \\ 0 & \text{o.w} \end{cases}$$

$$P(X < 1) = \int_0^1 f(x) dx$$

$$= \int_0^1 \frac{1}{2} e^{-\frac{1}{2}x} dx$$

$$= \frac{1}{2} \int_0^1 e^{-\frac{1}{2}x} dx$$

$$= \left(\frac{1}{2} \right) \left(\frac{e^{-\frac{1}{2}x}}{-\frac{1}{2}} \right) \Big|_0^1$$

$$= -e^{-\frac{1}{2}x} \Big|_0^1$$

$$= -e^{-\frac{1}{2}} + e^0$$

$$P(X < 1) = 1 - e^{-\frac{1}{2}} = 0.39346$$

$$\text{ii) } P(1 < X < 2) = \int_1^2 f(x) dx$$

$$= \int_1^2 \frac{1}{2} e^{-\frac{1}{2}x} dx$$

$$= \frac{1}{2} \int_1^2 e^{-\frac{1}{2}x} dx$$

$$= \left(\frac{1}{2} \right) \left(\frac{e^{-\frac{1}{2}x}}{-\frac{1}{2}} \right) \Big|_1^2$$

$$= -e^{-\frac{1}{2}x} \Big|_1^2$$

$$= -e^{-\frac{1}{2} \times 2} - (-e^{-\frac{1}{2} \times 1})$$

$$= -e^{-1} + e^{-\frac{1}{2}}$$

SOLUTION

$$P(1 < X < 2) = e^{-\frac{1}{2}} - e^{-1} \\ = 0.23865$$

$$\begin{aligned} \text{iii) } P(X > 2) &= \int_2^{\infty} f(x) dx \\ &= \int_2^{\infty} \frac{1}{2} e^{-\frac{1}{2}x} dx \\ &= \left(\frac{1}{2} \right) \frac{e^{-\frac{1}{2}x}}{\left(-\frac{1}{2} \right)} \bigg|_2^{\infty} \\ &= -e^{-\frac{1}{2}x} \bigg|_2^{\infty} \\ &= -e^{-\infty} - (-e^{-\frac{1}{2} \times 2}) \\ &= 0 + e^{-1} \\ &= 0.3678 \end{aligned}$$

QUESTIONS

The pdf of X is as given below

$$f(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & \text{o.w} \end{cases}$$

Find $P(X > 0.2)$ $P(1.2 < X < 1.5)$, Mean

SOLUTION

$$\text{Q } f(x) = 2e^{-2x} \quad x \geq 0 \\ = 0 \quad 0 < x$$

$$\text{i) } P(x > 0.2) = \int_{0.2}^{\infty} f(x) dx$$

$$= \int_{0.2}^{\infty} 2e^{-2x} dx$$

$$= 2 \int_{0.2}^{\infty} e^{-2x} dx$$

$$= 2 \left. \frac{e^{-2x}}{-2} \right|_{0.2}^{\infty}$$

$$= -e^{-2x} \Big|_{0.2}^{\infty}$$

$$= -e^{-\infty} - (-e^{-2 \times 0.2})$$

$$= 0 + e^{-0.4}$$

$$= 0.6703$$

$$\text{ii) } P(1.2 < x < 1.5) = \int_{1.2}^{1.5} f(x) dx$$

$$= \int_{1.2}^{1.5} 2e^{-2x} dx$$

$$= 2 \int_{1.2}^{1.5} e^{-2x} dx$$

$$= 2 \left. \frac{e^{-2x}}{-2} \right|_{1.2}^{1.5}$$

$$= -e^{-2x} \Big|_{1.2}^{1.5}$$

$$= -e^{-2(1.5)} - (-e^{-2(1.2)})$$

$$= -e^{-3} + e^{-2.4}$$

$$= 0.0409$$

iii) Mean

$$\therefore X \sim \exp(\lambda = 2)$$

$$\text{Mean} = \frac{1}{\lambda} = \frac{1}{2}$$

QUESTIONS

The life of a bulb is assumed to be exponentially distributed with mean life of 25 hours. Find the chance that life of bulb

- a. Exceeds 30 hours
- b. Is less than 35 hours

SOLUTION

Q X : Lifetime of a bulb

$$X \sim \exp(1)$$

$$f(x) = \begin{cases} 1e^{-1x} & x > 0 \\ 0 & 0 \leq x \end{cases}$$

Mean life of bulb = 25
i.e. $\frac{1}{\lambda} = 25$

$$\lambda = \frac{1}{25}$$

$$\therefore X \sim \exp\left(\lambda = \frac{1}{25}\right)$$

$$f(x) = \begin{cases} \frac{1}{25} e^{-\frac{1}{25}x} & x > 0 \\ 0 & 0 \leq x \end{cases}$$

a) $P(\text{Exceeds 30 hours})$

$$P(X > 30) = \int_{30}^{\infty} f(x) dx$$

$$= \int_{30}^{\infty} \frac{1}{25} e^{-\frac{1}{25}x} dx$$

$$= \frac{1}{25} \int_{30}^{\infty} e^{-\frac{1}{25}x} dx$$

SOLUTION

$$\begin{aligned} &= \left(\frac{1}{25}\right) \frac{e^{-\frac{1}{25}x}}{\left(-\frac{1}{25}\right)} \bigg|_{30}^{\infty} \\ &= -e^{-\frac{1}{25}x} \bigg|_{30}^{\infty} \\ &= -e^{-\frac{1}{25}(\infty)} - \left(-e^{-\frac{1}{25} \times 30}\right) \\ &= -e^{-\infty} + e^{-\frac{30}{25}} \\ &= 0 + e^{-\frac{30}{25}} \\ &= 0.30119 \end{aligned}$$

b) P(Less than 35 hours)

$$\begin{aligned} P(X < 35) &= \int_0^{35} f(x) dx \\ &= \int_0^{35} \frac{1}{25} e^{-\frac{1}{25}x} dx \\ &= \frac{1}{25} \frac{e^{-\frac{1}{25}x}}{\left(-\frac{1}{25}\right)} \bigg|_0^{35} \end{aligned}$$

$$\begin{aligned} &= -e^{-\frac{1}{25}x} \bigg|_0^{35} \\ &= -e^{-\frac{1}{25}(35)} - \left(-e^{-\frac{1}{25}(0)}\right) \\ &= -e^{-\frac{35}{25}} + e^0 \\ &= -e^{-\frac{35}{25}} + 1 \\ &= 0.7534 \end{aligned}$$

QUESTION

The lifetime of an electronic device is a random variable assuming the following pdf

$$f(x) = \begin{cases} 0.01e^{-0.01x} & x \geq 0 \\ 0 & \text{o.w} \end{cases}$$

What is the mean lifetime of the device

What is the Probability that Device will fail in the last 40 days

What is the probability that device will last at least 50 days

What is the probability that device will last at most 30 days

What is the probability that device will last between 40 to 50 days

SOLUTION

$$Q \quad f(x) = \begin{cases} 0.01 e^{-0.01x} & x \geq 0 \\ 0 & \text{o.w} \end{cases}$$

i) Mean lifetime of device
 $X \sim \exp(\lambda = 0.01)$

$$\therefore \text{Mean} = \frac{1}{\lambda} = \frac{1}{0.01} = 100$$

ii) $P(\text{Device will fail in first 40 days})$

$$= P(X < 40)$$

$$= \int_0^{40} f(x) dx$$

$$= \int_0^{40} 0.01 e^{-0.01x} dx$$

$$= 0.01 \frac{e^{-0.01x}}{-0.01} \Big|_0^{40}$$

$$= -e^{-0.01x} \Big|_0^{40}$$

$$= -e^{-0.01(40)} - (-e^{-0.01(0)})$$

$$= -e^{-0.01(40)} + 1$$

$$= 0.32967$$

iii) $P(\text{At least 50 days})$
 $P(X > 50)$

$$= \int_{50}^{\infty} f(x) dx$$

$$= \int_{50}^{\infty} 0.01 e^{-0.01x} dx$$

$$= 0.01 \frac{e^{-0.01x}}{-0.01} \Big|_{50}^{\infty}$$

$$= -e^{-0.01x} \Big|_{50}^{\infty}$$

$$= -e^{-0.01(\infty)}$$

$$= -e^{-0.01(\infty)} - (-e^{-0.01(50)})$$

$$= 0 + e^{-0.01(50)}$$

$$= 0.60653$$

SOLUTION

iv) P(Atmost 30 days)

$P(X < 30)$

$$= \int_0^{30} f(x) dx$$

$$= \int_0^{30} 0.01 e^{-0.01x} dx$$

$$= 0.01 \frac{e^{-0.01x}}{(-0.01)} \Big|_0^{30}$$

$$= -e^{-0.01x} \Big|_0^{30}$$

$$= -e^{-0.01(30)} - (-e^{-0.01(0)})$$

$$= -e^{-0.01(30)} + e^{-0}$$

$$= -e^{-0.01(30)} + 1$$

$$= 0.2591$$

v) Between 40 to 50 days

$P(40 < X < 50)$

$$= \int_{40}^{50} f(x) dx$$

$$= \int_{40}^{50} 0.01 e^{-0.01x} dx$$

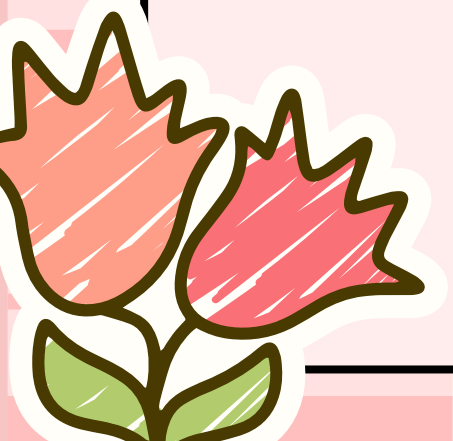
$$= 0.01 \frac{e^{-0.01x}}{(-0.01)} \Big|_{40}^{50}$$

$$= -e^{-0.01x} \Big|_{40}^{50}$$

$$= -e^{-0.01(50)} - (-e^{-0.01(40)})$$

$$= 0.06378$$

SOLUTION



QUESTIONS

$X \sim \text{Exp}(\lambda = 5)$; where λ is the mean of exponential distribution

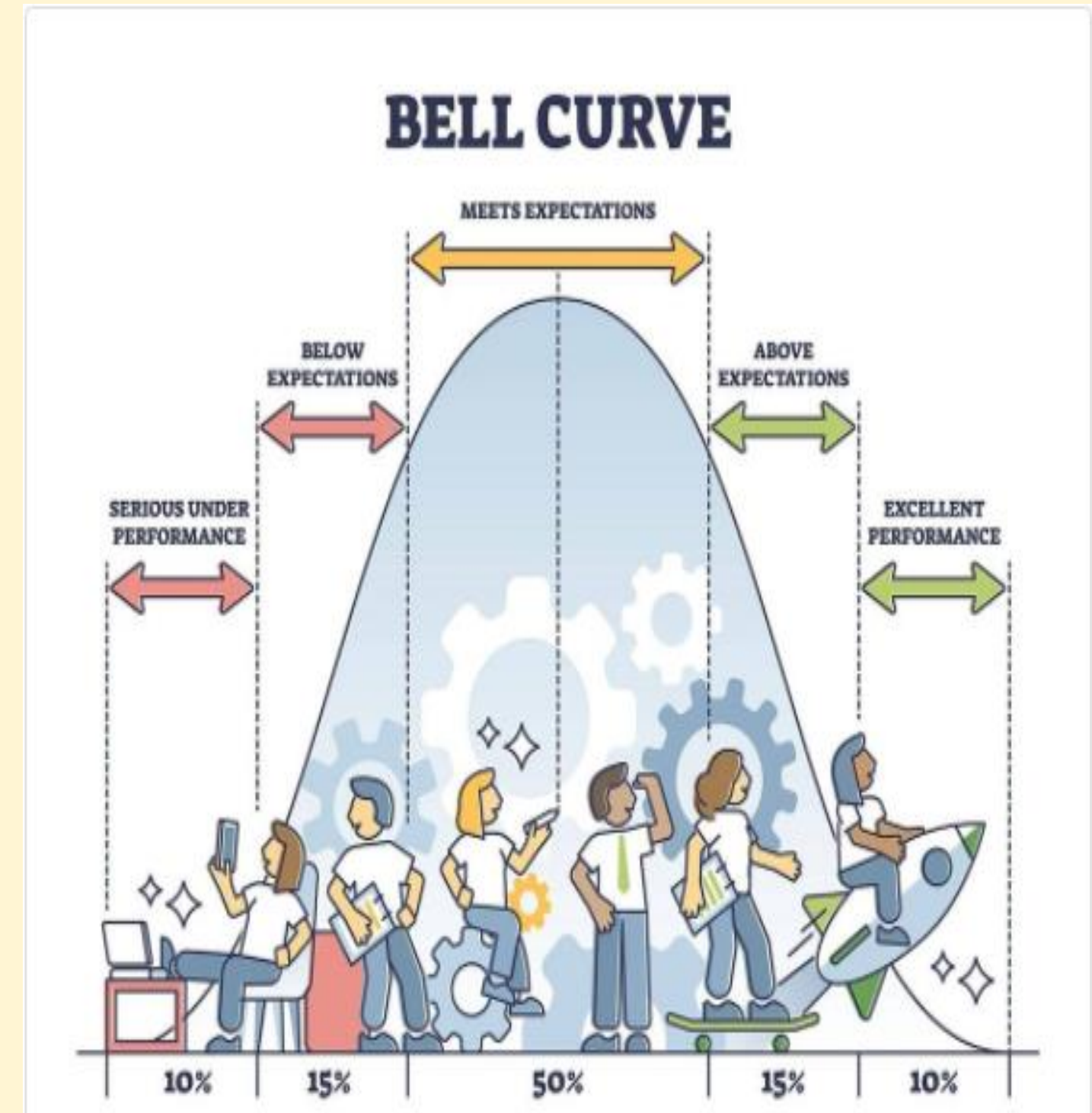
Find $P(X < 2)$, $P(X \geq 1.5)$, $P(1 < X < 2)$

Normal Distribution:

Also called Gaussian distribution, the most common distribution function for independent, randomly generated variables.

The graph of the normal distribution is characterized by two parameters:

1. the mean, or average, which is the maximum of the graph and about which the graph is always symmetric
2. The standard deviation, which determines the amount of dispersion away from the mean.



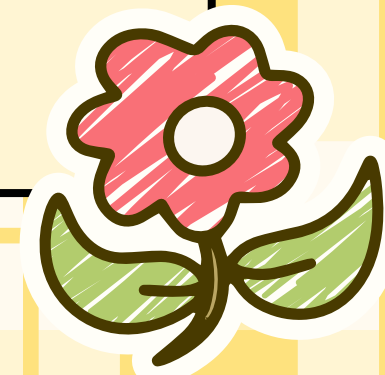
A random variable X is said to have a normal distribution with parameters μ (called "mean") and σ^2 (called "variance") if its density function is given by the probability law:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}, -\infty < x < \infty,$$
$$-\infty < \mu < \infty, \sigma > 0$$

Notation: $X \sim N(\mu, \sigma^2)$

If $X \sim N(\mu, \sigma^2)$ then $Z = \frac{x-\mu}{\sigma}$ is a standard normal variate with $E(Z) = 0$ and $Var(Z) = 1$ and we write $Z \sim N(0,1)$

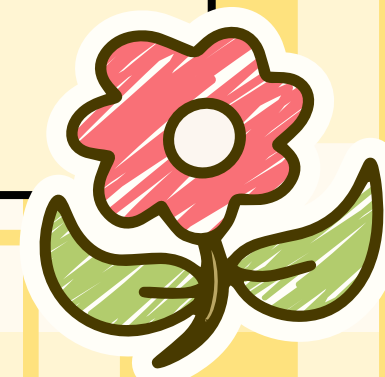
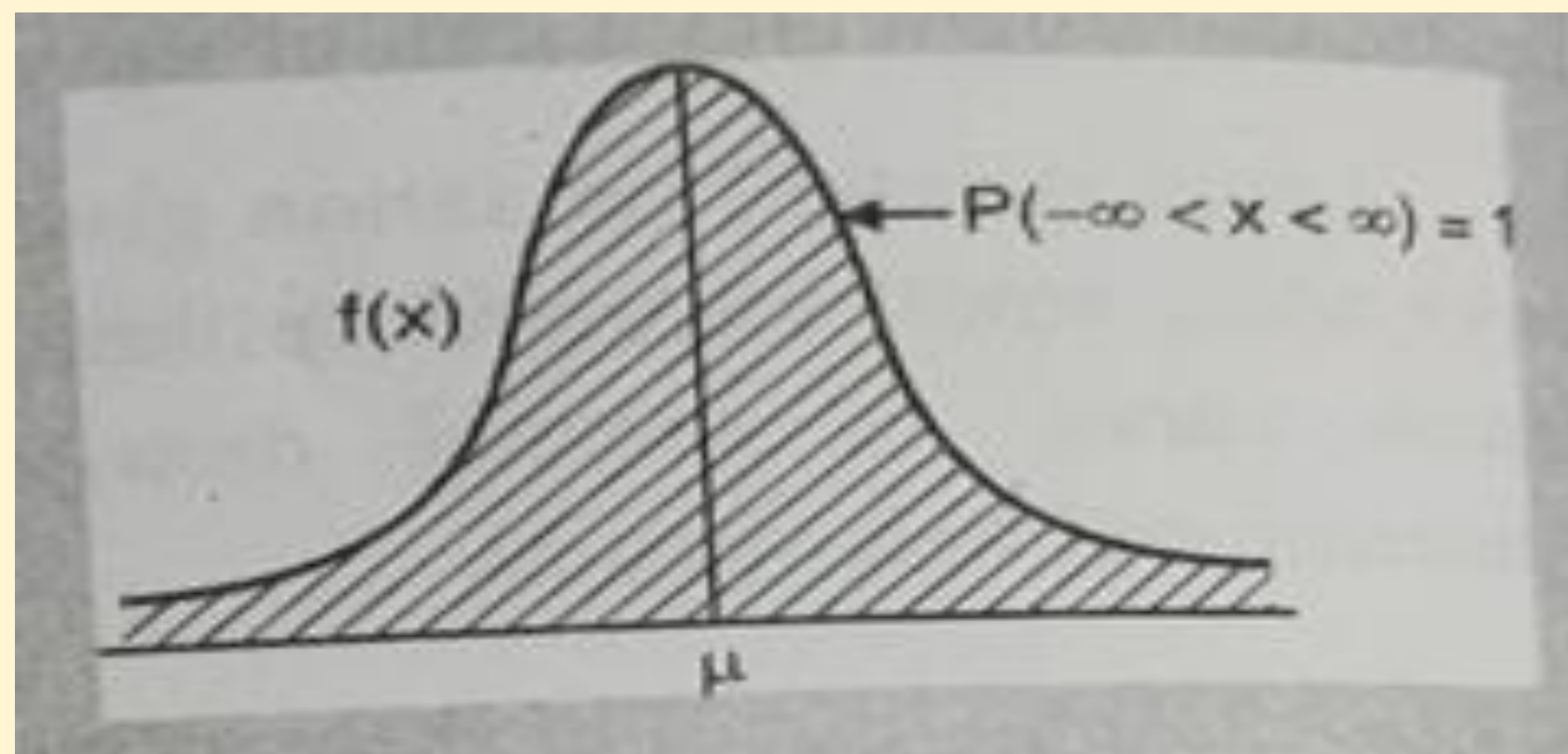
$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}, -\infty < x < \infty,$$



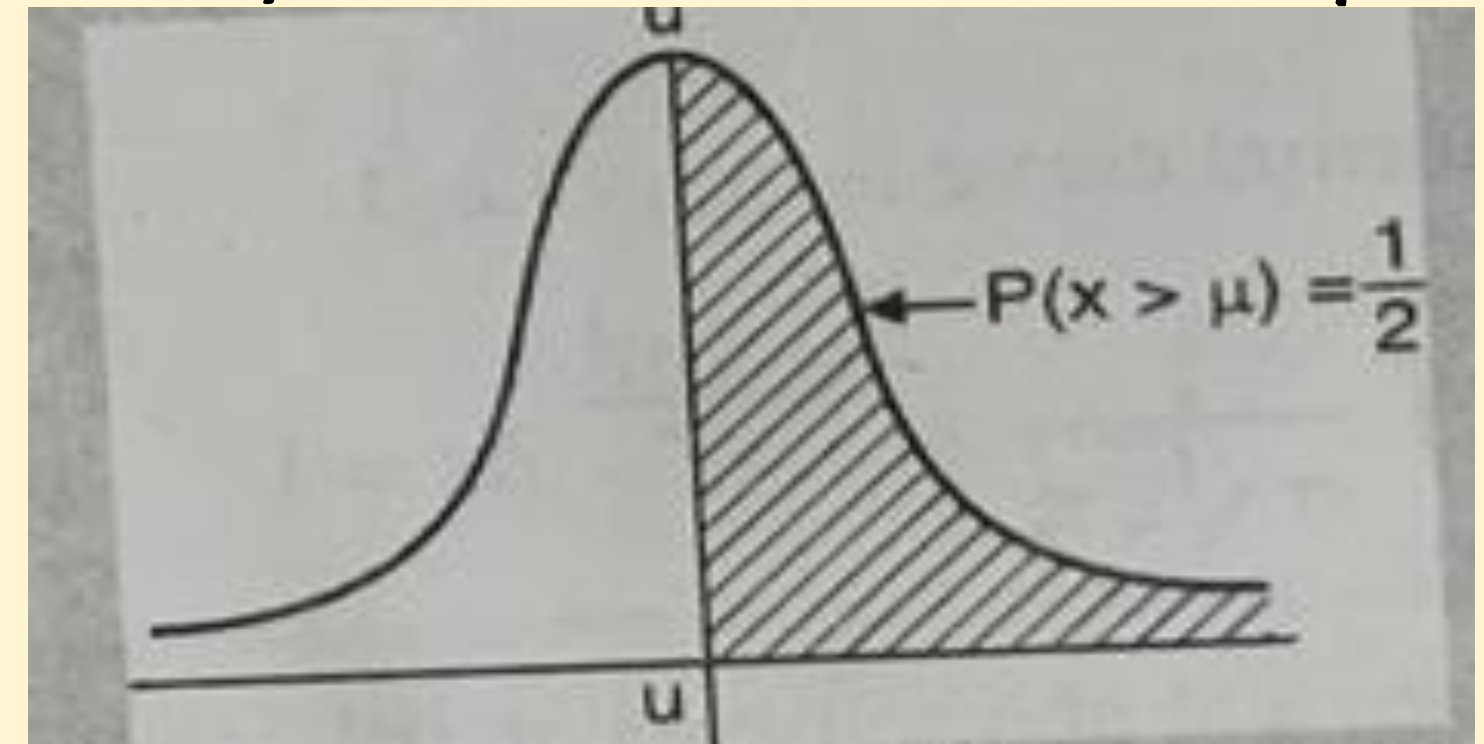
1. If $f(x)$ is a p.d.f of X ;

then $P(a < X < b) = \int_a^b f(x)dx$ i.e the area under the density curve $f(x)$ is bounded by X -axis and $X=a$, $X=b$

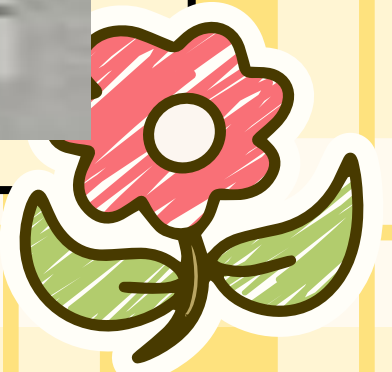
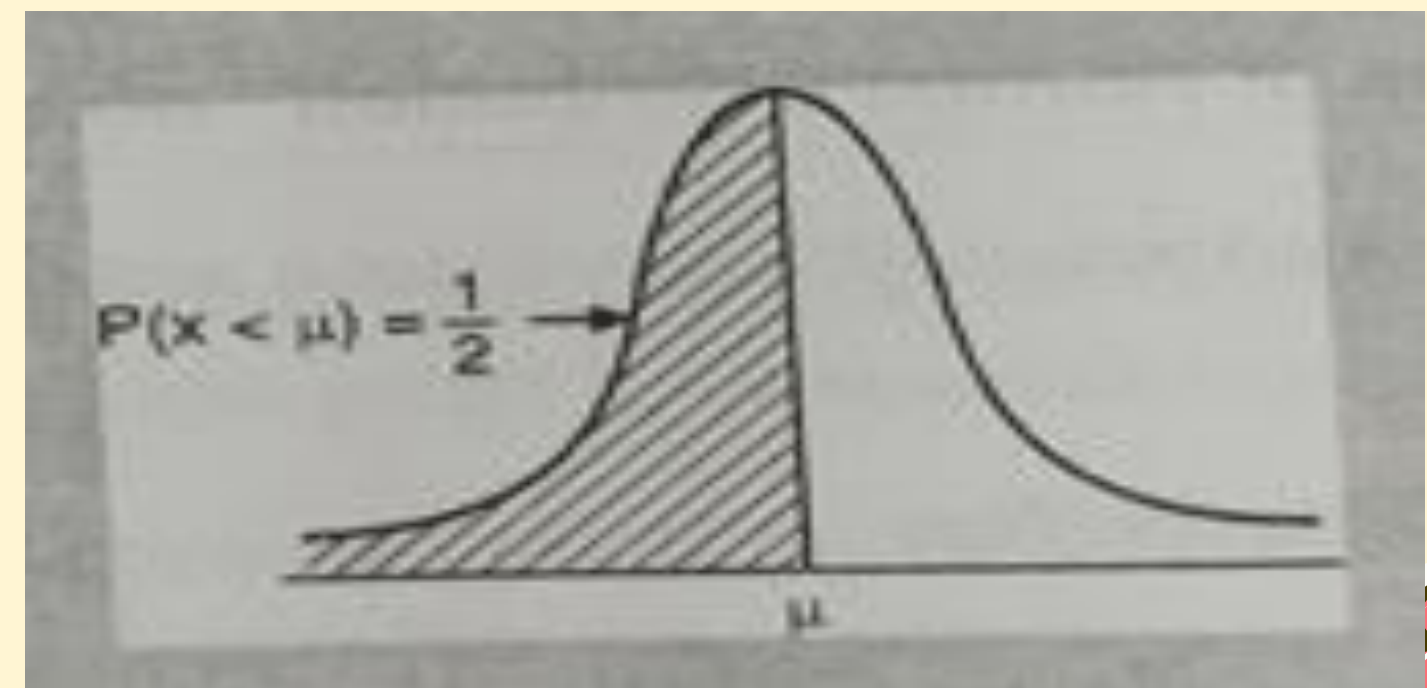
$$P(-\infty < x < \infty) = \int_{-\infty}^{\infty} f(x)dx = 1$$



2. $P(X > \mu) = \frac{1}{2}$ since the distribution is symmetric about mean μ

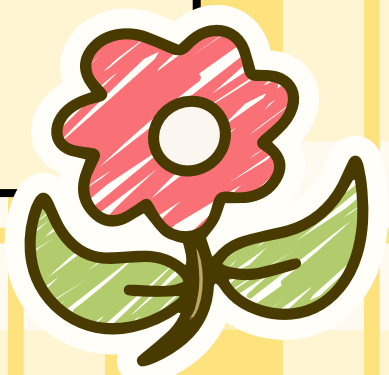


3. $P(X < \mu) = \frac{1}{2}$



Properties of Normal Distribution

1. In a normal distribution, the mean, median and mode are equal.(i.e., Mean = Median= Mode).
2. The total area under the curve should be equal to 1.
3. The normally distributed curve should be symmetric at the centre.
4. There should be exactly half of the values are to the right of the centre and exactly half of the values are to the left of the centre.
5. The normal distribution should be defined by the mean and standard deviation.
6. The normal distribution curve must have only one peak. (i.e., Unimodal)
7. The curve approaches the x-axis, but it never touches, and it extends farther away from the mean.



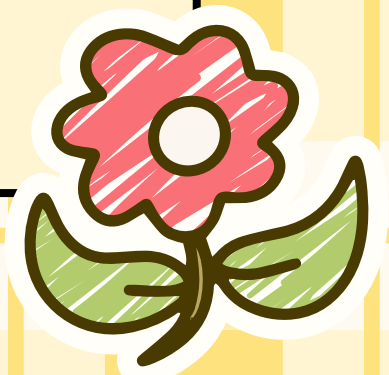
Gamma Distribution

The **gamma distribution** term is mostly used as a distribution which is defined as two parameters – shape parameter and inverse scale parameter, having continuous probability distributions. It is related to the normal distribution, [exponential distribution](#), chi-squared distribution and Erlang distribution. 'Γ' denotes the gamma function.

Gamma distributions have two free parameters, named as alpha (α) and beta (β), where;

- α = Shape parameter
- β = Rate parameter (the reciprocal of the scale parameter)

It is characterized by mean $\mu = \alpha\beta$ and variance $\sigma^2 = \alpha\beta^2$

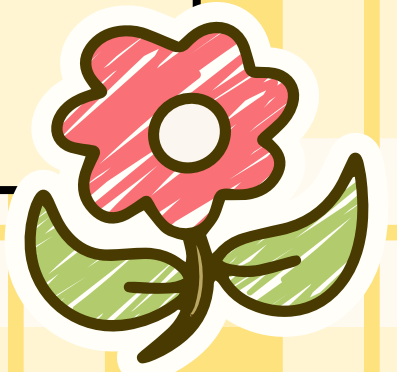
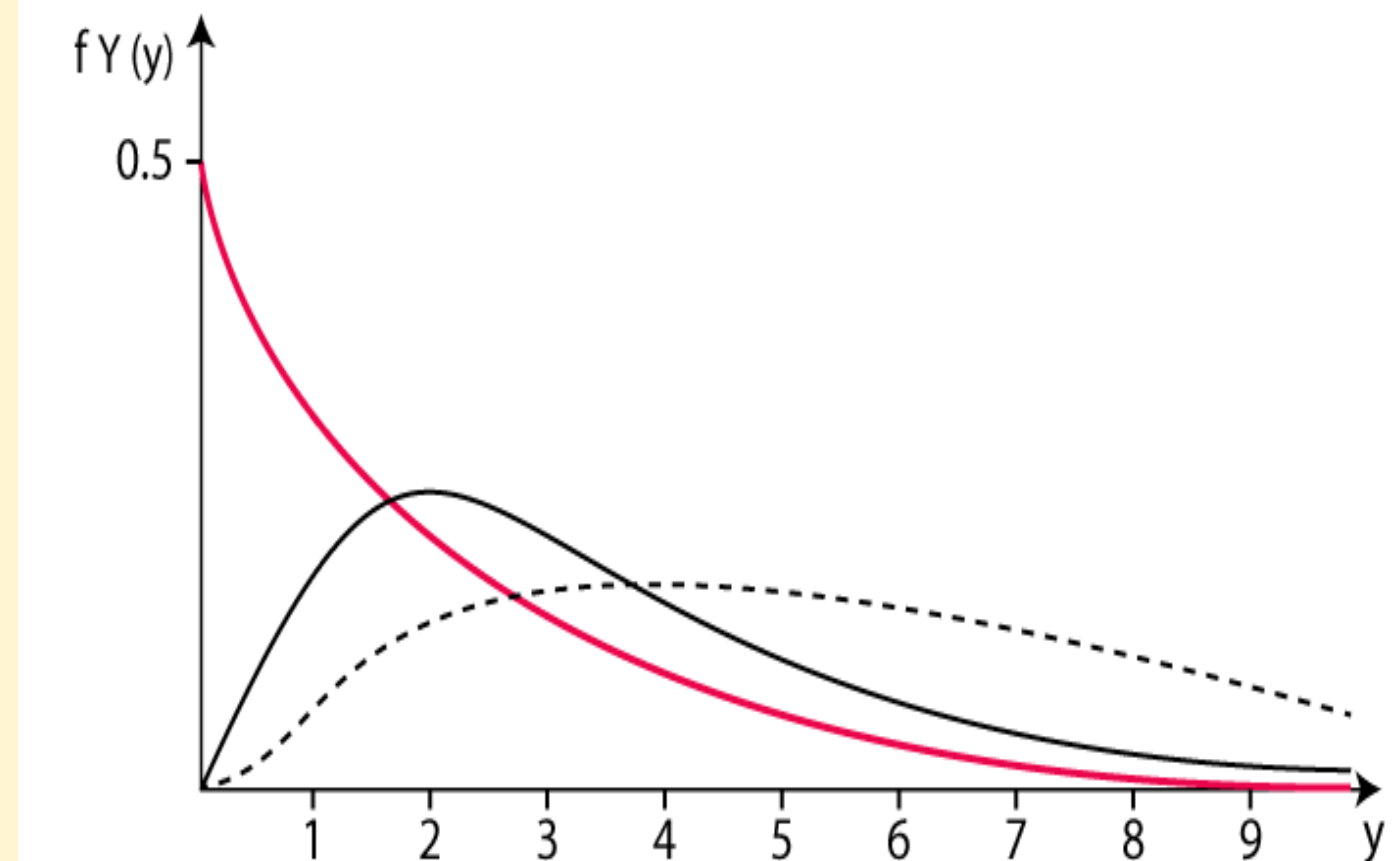


$$f(x|\alpha, \beta) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

The parameters of the gamma distribution define the shape of the graph. Shape parameter α and rate parameter β are both greater than 1.

When $\alpha = 1$, this becomes the exponential distribution

When $\beta = 1$ this becomes the standard gamma distribution



Gamma Distribution Properties

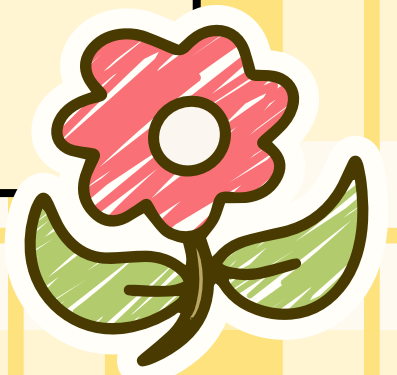
$$1. \Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx;$$

$$2. \int_0^{\infty} x^{\alpha-1} e^{-\lambda x} dx = \frac{\Gamma(\alpha)}{\lambda^{\alpha}}, \quad \text{for } \lambda > 0;$$

$$3. \Gamma(\alpha + 1) = \alpha \Gamma(\alpha);$$

$$4. \Gamma(n) = (n-1)!, \text{ for } n = 1, 2, 3, \dots;$$

$$5. \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}.$$



Gamma Distribution Mean:

There are two ways to determine the gamma distribution mean

1. Directly
2. Expanding the moment generation function

It is also known as the Expected value of Gamma Distribution.

Mean of Gamma

$$f(x) = \frac{e^{-x} x^{p-1}}{\Gamma p} \quad x \geq 0$$

$$= 0 \quad 0 < x$$

$$\therefore E(X) = \int_0^{\infty} x f(x) dx$$

$$= \int_0^{\infty} x \cdot \frac{e^{-x} x^{p-1}}{\Gamma p} dx$$

$$= \frac{1}{\Gamma p} \int_0^{\infty} x e^{-x} x^{p-1} dx$$

$$= \frac{1}{\Gamma p} \int_0^{\infty} e^{-x} x^{p-1+1} dx$$

$$= \frac{1}{\Gamma p} \int_0^{\infty} e^{-x} x^p dx \quad \text{--- (1)}$$

Using Property.

$$\int_0^{\infty} x^{\alpha-1} e^{-\lambda x} dx = \frac{\Gamma \alpha}{\lambda^{\alpha}} \quad \text{--- (2)}$$

comparing $e^{-x} x^p$ with $x^{\alpha-1} e^{-\lambda x}$ we get

$$\therefore \lambda = 1 \quad p = \alpha - 1$$

$$\lambda = 1; \alpha = p + 1$$

Substituting in (2) we get

$$\therefore E(X) = \frac{1}{\Gamma p} \frac{\Gamma(p+1)}{1^{p+1}}$$

$$= \frac{\Gamma(p+1)}{\Gamma p}$$

$\therefore$ Using Property $\Gamma n = (n-1)!$

$$\therefore E(X) = \frac{(p+1-1)!}{(p-1)!}$$

$$= \frac{p!}{(p-1)!}$$

$$= \frac{p!}{(p-1)!}$$

$$= \frac{p(p-1)!}{(p-1)!}$$

$$\boxed{E(X) = p}$$

Gamma Distribution Variance:

Variance of Gamma

$$V(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = \int_0^{\infty} x^2 f(x) dx$$

$$\because f(x) = \frac{e^{-x} x^{p-1}}{\Gamma(p)}$$

$$\therefore E(X^2) = \int_0^{\infty} x^2 \frac{e^{-x} x^{p-1}}{\Gamma(p)} dx$$

$$= \frac{1}{\Gamma(p)} \int_0^{\infty} x^2 e^{-x} x^{p-1} dx$$

$$= \frac{1}{\Gamma(p)} \int_0^{\infty} e^{-x} x^{p-1+2} dx$$

$$= \frac{1}{\Gamma(p)} \int_0^{\infty} e^{-x} x^{p+1} dx \quad \text{--- (1)}$$

$$\text{Using Property } \int_0^{\infty} x^{\alpha-1} e^{-\lambda x} dx = \frac{\Gamma(\alpha)}{\lambda^{\alpha}}$$

comparing the integrand^{term} in (1) with the property

$$\begin{array}{l} e^{-x} x^{p+1} \text{ with } x^{\alpha-1} e^{-\lambda x} \\ \text{we get } \lambda = 1 \quad p+1 = \alpha-1 \\ \therefore \lambda = 1 \quad \alpha = p+2 \end{array}$$

Now using Property in (1)

$$\begin{aligned} E(X^2) &= \frac{1}{\Gamma(p)} \frac{\Gamma(p+2)}{1^{p+2}} \\ &= \frac{\Gamma(p+2)}{\Gamma(p)} \end{aligned}$$

Using $\Gamma(n) = (n-1)!$

$$= \frac{(p+2-1)!}{(p-1)!}$$

$$= \frac{(p+1)!}{(p-1)!}$$

$$= \frac{(p+1) p (p-1)!}{(p-1)!}$$

$$E(X^2) = p(p+1)$$

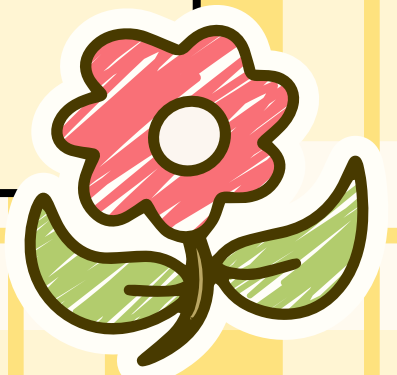
$$\therefore E(X) = p$$

$$\begin{aligned} \therefore V(X) &= E(X^2) - [E(X)]^2 \\ &= p(p+1) - p^2 \end{aligned}$$

$$\boxed{V(X) = \frac{p^2 + p - p^2}{p}} \quad \text{--- (2)}$$

Find the value of the following functions

1. $\Gamma(2.5)$
2. $\Gamma(3.5)$
3. $2\Gamma(1.5)$
4. $4\Gamma(0.5)$
5. $\Gamma(5.5)$
6. $\Gamma(5)$
7. $\Gamma(4)$



1.

i) Gamma (2.5)

$$\Gamma(2.5) = \Gamma\left(\frac{25}{10}\right) = \Gamma\left(\frac{5}{2}\right)$$

Using Property $\Gamma(\alpha) = \alpha - 1 \Gamma(\alpha - 1)$

$$\therefore \Gamma\left(\frac{5}{2}\right) = \left(\frac{5}{2} - 1\right) \Gamma\left(\frac{5}{2} - 1\right)$$

$$= \frac{3}{2} \Gamma\left(\frac{3}{2}\right)$$

$$= \frac{3}{2} \times \left(\frac{3}{2} - 1\right) \Gamma\left(\frac{3}{2} - 1\right)$$

$$= \frac{3}{2} \times \frac{1}{2} \Gamma\left(\frac{1}{2}\right)$$

$$= \frac{3}{2} \times \frac{1}{2} \times \sqrt{\pi} \quad \because \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$

$$= \frac{3\sqrt{\pi}}{4}$$

2.

Gamma (3.5)

$$\Gamma(3.5) = \Gamma\left(\frac{35}{10}\right) = \Gamma\left(\frac{7}{2}\right)$$

Using Property $\Gamma(\alpha) = \alpha - 1 \Gamma(\alpha - 1)$

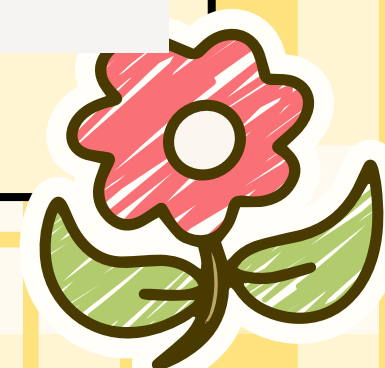
$$\Gamma\left(\frac{7}{2}\right) = \frac{5}{2} \cdot \Gamma\left(\frac{5}{2}\right) \quad (\text{u})$$

$$= \frac{5}{2} \cdot \frac{3}{2} \cdot \Gamma\left(\frac{3}{2}\right)$$

$$= \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right)$$

$$= \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}$$

$$= \frac{15}{8} \sqrt{\pi}$$



3.

2 Gamma(1.5)

$$2 \sqrt{1.5} = 2 \sqrt{\frac{15}{10}} = 2 \sqrt{\frac{3}{2}}$$

Using Property $\sqrt{\alpha} = \alpha-1 \sqrt{\alpha-1}$

$$\therefore 2 \sqrt{\frac{3}{2}} = 2 \left(\frac{3}{2} - 1 \right) \sqrt{\frac{3}{2} - 1}$$

$$= 2 \left(\frac{1}{2} \right) \sqrt{\frac{1}{2}}$$

$$= 2 \times \frac{1}{2} \times \sqrt{\frac{1}{2}}$$

$$= \sqrt{\frac{1}{2}}$$

$$\because \sqrt{\frac{1}{2}} = \frac{\sqrt{\pi}}{2}$$

4.

4 Gamma(0.5)

$$4 \sqrt{0.5} = 4 \sqrt{\frac{5}{10}} = 4 \sqrt{\frac{1}{2}}$$

$$= 4 \sqrt{\frac{1}{2}}$$

$$\because \sqrt{\frac{1}{2}} = \frac{\sqrt{\pi}}{2}$$

5.

5 Gamma(5.5)

$$\sqrt{5.5} = \sqrt{\frac{55}{10}} = \sqrt{\frac{11}{2}}$$

Using Property $\sqrt{\alpha} = \alpha-1 \sqrt{\alpha-1}$

$$\therefore \sqrt{\frac{11}{2}} = \left(\frac{11}{2} - 1 \right) \sqrt{\frac{11}{2} - 1}$$

$$= \frac{9}{2} \sqrt{\frac{9}{2}}$$

$$= \frac{9}{2} \left(\frac{9}{2} - 1 \right) \sqrt{\frac{9}{2} - 1}$$

$$= \frac{9}{2} \times \frac{7}{2} \sqrt{\frac{7}{2}}$$

$$= \frac{9}{2} \times \frac{7}{2} \times \frac{5}{2} \sqrt{\frac{5}{2}}$$

$$= \frac{9}{2} \times \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \sqrt{\frac{3}{2}}$$

$$= \frac{9}{2} \times \frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \sqrt{\frac{1}{2}}$$

$$= \frac{9 \times 7 \times 5 \times 3 \times 1}{2^5} \sqrt{\frac{1}{2}} \quad \because \sqrt{\frac{1}{2}} = \frac{\sqrt{\pi}}{2}$$

$$= \frac{945 \sqrt{\pi}}{32}$$



6.

Gamma(5)

$\Gamma 5$

Using $\Gamma n = (n-1)!$

$$\begin{aligned}\therefore \Gamma 5 &= (5-1)! \\ &= 4! \\ &= 4 \times 3 \times 2 \times 1 \\ &= 24\end{aligned}$$

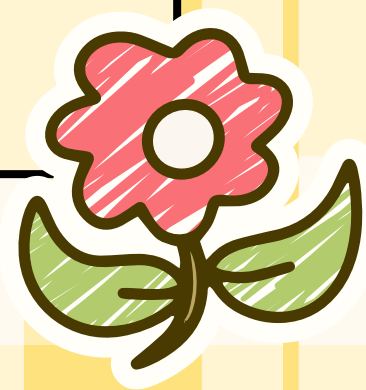
7.

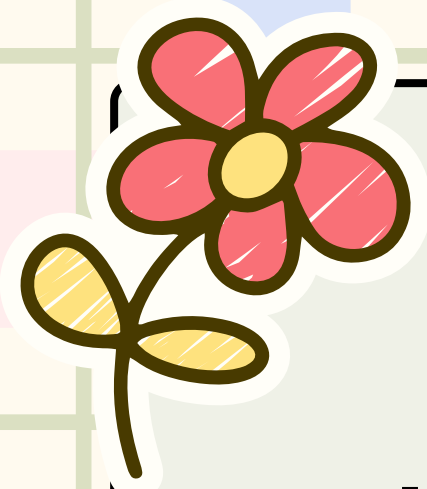
Gamma(4)

$\Gamma 4$

Using $\Gamma n = (n-1)!$

$$\begin{aligned}\therefore \Gamma 4 &= (4-1)! \\ &= 3! \\ &= 3 \times 2 \times 1 \\ &= 6\end{aligned}$$

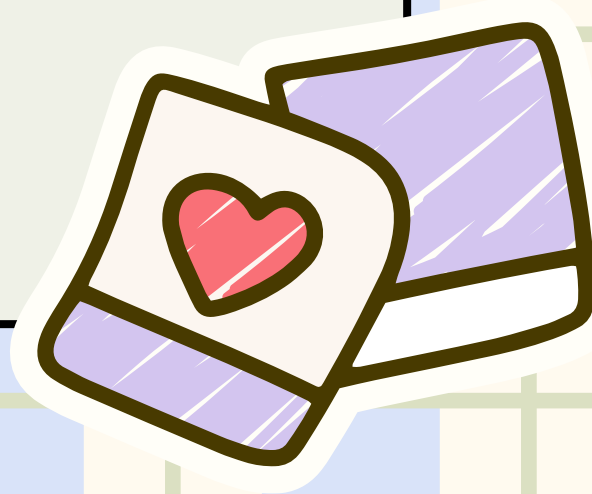


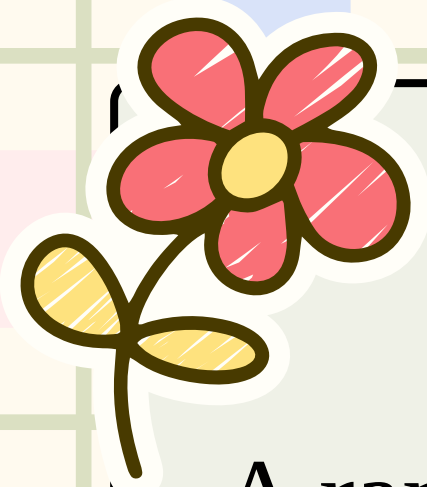


Beta Distribution



1. The beta distribution is a family of continuous probability distributions set on the interval $[0, 1]$ having two positive shape parameters, expressed by α and β .
2. These two parameters appear as exponents of the random variable and manage the shape of the distribution.
3. Usually, the basic distribution is known as the Beta distribution of its first kind, and prime beta distribution is called for its second kind.
4. The most common use of this distribution is to model the uncertainty about the probability of success of a random experiment.





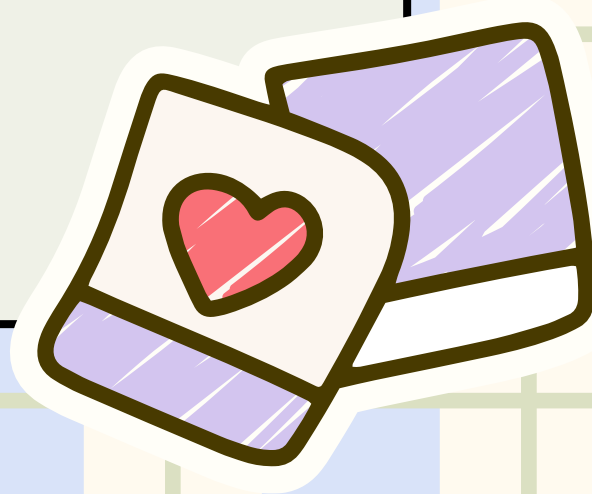
Type 1 Beta Distribution

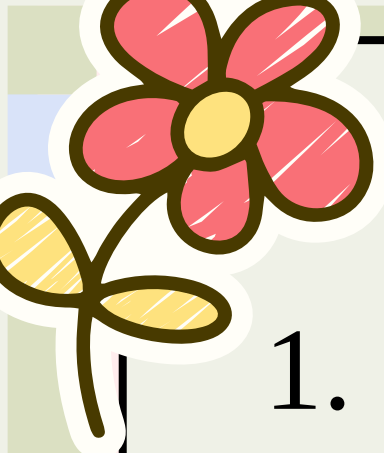


A random variable X is said to follow beta distribution of first kind with parameters $m > 0$ and $n > 0$, if its probability density function is given by

$$f(x) = \begin{cases} \frac{1}{\beta(m, n)} x^{m-1} (1-x)^{n-1}, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

The random variable X is known as beta variate of first kind and can be expressed as $X \sim \beta_1(m, n)$





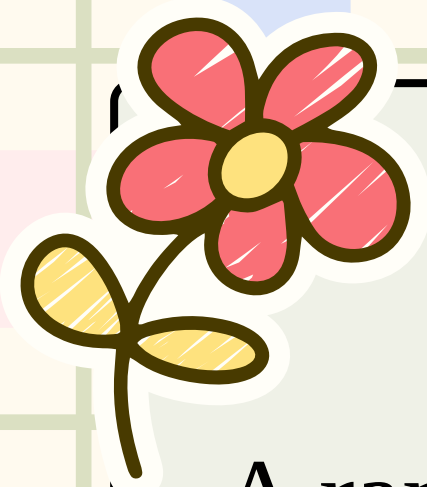
Properties of Beta Type 1 Distribution

1. The Beta Type 1 distribution is limited to values between 0 and 1, making it perfect for modeling probabilities.
2. It is determined by two parameters, α and β , both positive numbers, which determine its shape and characteristics.
3. The skewness and kurtosis of the Beta Type 1 distribution depend on the values of its shape parameters α and β .
4. The mean and Variance of Beta 1 distribution is :

$$\text{Mean} = \frac{\alpha}{\alpha + \beta} \quad \text{Variance} = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

5. It is used in various fields, including reliability analysis, survival analysis etc. It is particularly useful when modeling data that are strictly positive and have a skewed distribution.





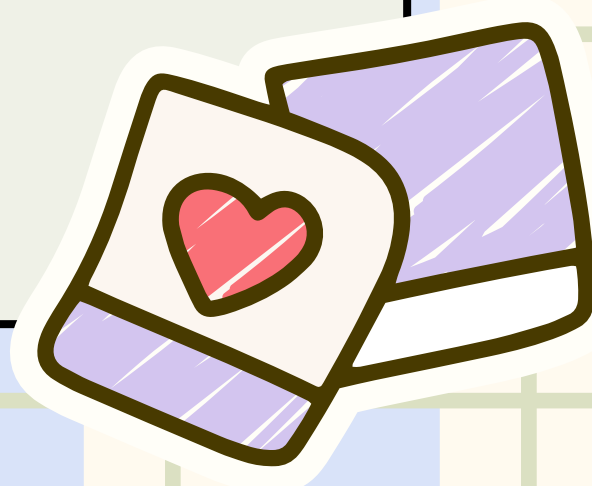
Type 1 Beta Distribution

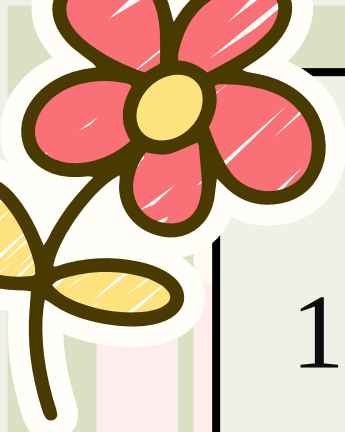


A random variable X is said to follow beta distribution of second kind with parameters $m > 0$, $n > 0$ if its probability density function is given by

$$f(x) = \begin{cases} \frac{x^{m-1}}{\beta(m, n)(1+x)^{m+n}}, & 0 < x < \infty \\ 0, & \text{elsewhere} \end{cases}$$

If X is a beta variate of second kind with parameters $m > 0$, $n > 0$, then it is expressed as $X \sim \beta_2(m, n)$





Properties of Beta Type II Distribution

1. The distribution is defined on the interval $[0,1]$, meaning it assigns probabilities only to values within this range.
2. The shape parameters α and β control the shape of the distribution. Larger values of α or β lead to more concentrated distributions around the mode.
3. The mean (μ) and variance (σ^2) of the Beta Type 2 distribution are given by: $\mu = \frac{\alpha}{\alpha+\beta}$, $\sigma^2 = \frac{\alpha\beta}{[(\alpha+\beta)^2(\alpha+\beta+1)]}$
4. The skewness and kurtosis of the distribution depend on the shape parameters α and β . However, it generally has finite moments and can be symmetric or skewed, depending on the values of α and β .
5. The Beta Type 2 distribution is related to the Beta distribution. When $\alpha = \beta = 1$, it reduces to the uniform distribution. It is also a special case of the generalized beta distribution.

